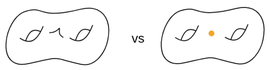
Obstructions to the realization of subgroups of mapping classes by diffeomorphisms

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Introduction

Given a compact orientable surface S_g and a finite set $\mathbf{z} = \{z_1, \dots, z_k\}$ we obtain the punctured surface $S_g^k := S_g \setminus \mathbf{z}$ or the marked surface $(S_g, \mathbf{z})$.



Their **mapping class groups** are isomorphic

$$\mathrm{Mcg}(S_g^k) := \frac{\mathrm{Diff}^+(S_g^k)}{\mathrm{Diff}_0(S_g^k)} \cong \mathrm{Mcg}(S_g, \mathbf{z}) := \frac{\mathrm{Diff}^+(S_g, \mathbf{z})}{\mathrm{Diff}_0(S_g, \mathbf{z})}$$

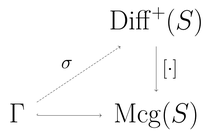
where Diff_0 is the path-connected component (normal subgroup) of the identity.

The group $\mathrm{Diff}^+(S)$ **acts naturally** on S. The question is:

Is there a natural action of $\mathrm{Mcg}(S)$ on the surface S as well?

The realization problem

Which subgroups $\Gamma \leq \mathrm{Mcg}(S)$ admit a **lift** to the diffeomorphism group $\mathrm{Diff}^+(S)$. That is a group homomorphism $\sigma : \Gamma \to \mathrm{Diff}^+(S)$ that makes the diagram



commute?

Answers for closed surfaces

Positive answers. Subgroups of $\mathrm{Mcg}(S_g)$ that have been proven to admit a lift to $\mathrm{Diff}^+(S_g)$ are all the following:

free subgroups, *finite cyclic subgroups* (Nielsen, 1943), *finite subgroups* (Kerckhoff, 1983), and *abelian subgroups* (Birman-Lubotzky-McCarthy, 1983).

Negative answers. For genus $g \geq 5$, Morita proved that $\mathrm{Mcg}(S_g)$ does not admit such a lift [1].

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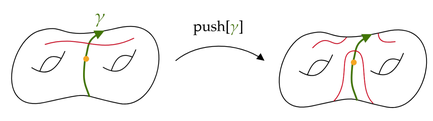
Non-lifting for punctured surfaces.

Theorem. (Bestvina, Church and Souto, [2].) Given a genus $g \geq 2$ compact orientable surface S_g and a set of $k \geq 1$ marked points $\mathbf{z} = \{z_1, \dots, z_k\}$

- finite index subgroups of the *point-pushing subgroup* $\mathbf{push}(\pi_1(S_g, z_1)) \leq \mathrm{Mcg}(S_g, z_1)$ do not admit a lift to $\mathrm{Diff}^+(S_g, z_1)$;
- finite index subgroups of $\mathrm{Mcg}(S_g, \mathbf{z})$ do not admit a lift to $\mathrm{Diff}^+(S_g, \mathbf{z})$.

Point-pushing map

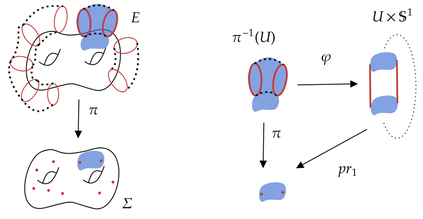
There is a monomorphism $\mathbf{push} : \pi_1(S_g, z) \hookrightarrow \mathrm{Mcg}(S_g, z)$ described by the following figure.



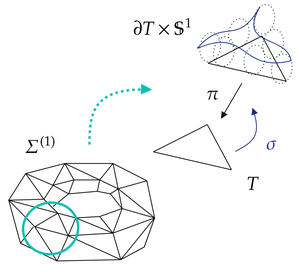
Its image $\mathbf{push}(\pi_1(S_g, z)) \cong \pi_1(S_g, z)$ is known as the *point-pushing subgroup*.

Euler class for $\mathbb{S}^1$-bundles

A **circle bundle** (or **$\mathbb{S}^1$-bundle**) over a surface is given by a surjection $\pi : E \to S_g$ whose fibers are homeomorphic to $\mathbb{S}^1$ and which admits local trivializations.

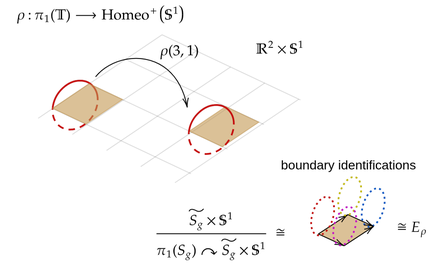


The **Euler class** of an **orientable** $\mathbb{S}^1$-bundle $E \to S_g$ is a characteristic class $e(E) = [c] \in H^2(S_g, \mathbb{Z})$ given by a cocycle, such that $c(T) = \deg(\text{map in the figure})$.



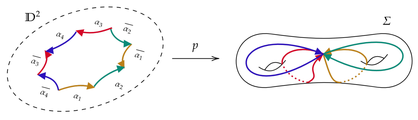
Flat circle bundles

Given a representation $\rho : \pi_1(S_g, z) \to \mathrm{Homeo}^+(\mathbb{S}^1)$, we produce a **diagonal** action $\pi_1(S_g, z) \curvearrowright \widetilde{S}_g \times \mathbb{S}^1$, whose quotient is an $\mathbb{S}^1$-bundle E_ρ over S_g.



In the first coordinate of $\widetilde{S}_g \times \mathbb{S}^1$ the action is described by **deck transformations** and in the second by ρ.

Remark. When $g \geq 2$ the universal cover of S_g admits a hyperbolic geometry, so the deck transformations can be realized by hyperbolic isometries.



Milnor-Wood inequality

Milnor gave a formula to compute the Euler class for flat bundles [3]. Later in [4], Wood proposed the upper bound for the Euler number of the flat bundle associated to $\rho : \pi_1(S_g, z) \to \mathrm{Homeo}^+(\mathbb{S}^1)$

$$|e(E_\rho)[S_g]| \leq \begin{cases} 2g - 2 = -\chi(S_g) & \text{if } g \geq 1, \\ 0 & \text{if } g = 0. \end{cases}$$

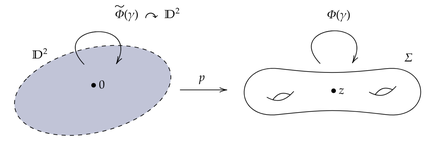
Moreover, if every element of $\rho(\pi_1(S_g, z))$ commutes with the antipodal map $\mathbf{a}(x) = -x$, then we get a better bound

$$|e(E_\rho)[S_g]| \leq \begin{cases} g - 1 = -\frac{1}{2}\chi(S_g) & \text{if } g \geq 1, \\ 0 & \text{if } g = 0. \end{cases}$$

Proof by contradiction

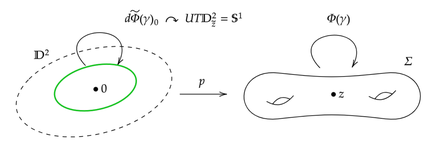
Suppose that **the point-pushing map lifts** to some homomorphism $\phi : \pi_1(S_g, z) \to \mathrm{Diff}^+(S_g, z)$.

(A) After lifting to the universal cover we obtain a representation $\widetilde{\phi} : \pi_1(S_g, z) \to \mathrm{Diff}^+(\mathbb{D}^2, 0)$.

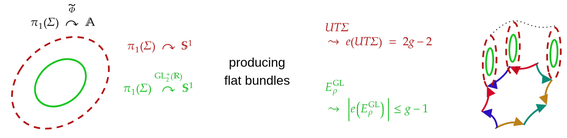


This action extends to $\partial_\infty \mathbb{D}^2$, where it agrees with the action of $\pi_1(S_g, z)$ by **deck transformations**.

(B) There is a **linear action** on the unit tangent circle given by $\pi_1(S_g, z) \xrightarrow{d\widetilde{\phi}_0} \mathrm{GL}_2^+(\mathbb{R}^2) \xrightarrow{\text{normalization}} \mathrm{Homeo}^+(\mathbb{S}^1)$.



(C) We combine both actions into a compactification of $\mathbb{D}^2 \setminus \{0\}$ to the annulus $\mathbb{A}$. This produces an $\mathbb{A}$-bundle over S_g.



A retraction of the $\mathbb{A}$-bundle to each $\mathbb{S}^1$-bundle yields $e(E_1) = e(E_2)$. This contradicts the Milnor-Wood inequality

$$|e(E_1)[S_g]| \leq g - 1 < 2g - 2 = e(E_2)[S_g].$$

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